

The Massive Interior: Ultraviolet Safety, Causality with Tails, and What Survives Mass

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Abstract. The mass paper installed the marriage mechanism and named cone compatibility as its first hurdle: the entire spacetime derivation of this series ran through *massless* structure, and mass had to be shown to live consistently in the world that structure derived. This note executes the compatibility architecture with three verifications. **(1) Ultraviolet safety.** The cone identification was an ultraviolet statement — leading singularities did the work — and mass is a relevant perturbation, invisible exactly there: the massive kernel's leading short-distance singularity is the massless one, verified by $R(x) = x^2 \partial_x^2 K_0(mx) \rightarrow 1$ (1.0589, 1.0467, 1.0056, 1.0004 at $x = 1.0, 0.5, 0.1, 0.02$), with the predicted $O(m^2 x^2 \log)$ correction visible in the approach. The derivation of spacetime is untouched by the mass that will inhabit it. **(2) Causality kept, Huygens lost — and the loss measured.** The massive commutator vanishes at spacelike separation (1.4×10^{-5} , regulator level, against interior values of order one): microcausality — the *derived axiom* of the interior-net paper — survives. What is lost is Huygens sharpness, which was never an axiom: the commutator grows an interior tail, matched quantitatively to its closed form, 0.332234 against $\frac{1}{2} J_0(m\sqrt{t^2 - x^2}) = 0.332459$. The Haag–Kastler structure crosses the frontier intact. **(3)** The massive wedge is thermal at $\beta = 2\pi$. ****** The Wightman function along the boost orbit, built from the K_0 integral with the $i\delta$ prescription (whose shift contributes the *exactly known* factor $e^{-(2\pi-2\delta)\omega}$), satisfies detailed balance across three and a half decades — measured ratios from 2.27×10^{-1} down to 8.80×10^{-5} tracking the prediction at 2–6% with no fitted parameters. Bisognano–Wichmann holds for the massive theory, so the wedge modular Hamiltonians, the stress-tensor first law, the relative-entropy machinery, and the equation-of-state derivation of gravity are all *mass-robust*. What is honestly massless-specific: the generator-diagonal cone data (the cone is the derivation scaffold; massive dynamics lives in the interior net), the conformal interval flows (the massive diamond modular Hamiltonian is the known-hard frontier, cited to Longo–Morsella), and exact scale symmetry (mass is a relevant coupling, entering the same dimension bookkeeping the covariance audit established — now in the infrared, where it is allowed). The architecture stands: **spacetime derived in the ultraviolet by the massless scaffold; mass installed in the infrared by the marriage and its condensate; consistency verified where the two meet.**

1. The division of labor

The hurdle, stated sharply: Huygens propagation — massless sharpness — is what made the cone data generator-diagonal and the reconstruction formulas exact, and mass destroys Huygens. The resolution is a division of labor the verifications below make quantitative. The derivation of spacetime (cone identification, Lorentz invariance as output, the interior net) is ultraviolet and kinematic; mass is a relevant perturbation, dominant in the infrared and invisible in the ultraviolet. The massive theory is therefore *installed in the interior* —

the bulk net the derivation delivered — while the massless cone remains the scaffold that derived the arena. Nothing in this is conjectural at the free-massive level: the massive free field on the derived Minkowski space is standard; what required proof was that the archive's load-bearing structures survive, and which do not.

2. Ultraviolet safety

Claim. The massive two-point kernel's leading short-distance singularity coincides with the massless kernel; corrections are $O(m^2 x^2 \log mx)$. **Verification.** $R(x) = x^2 \partial_x^2 K_0(mx) = x^2 m^2 (K_0(mx) + K_1(mx)/mx)$, computed from the Bessel integral representations: $R = 1.0589, 1.0467, 1.0056, 1.0004$ at $x = 1.0, 0.5, 0.1, 0.02$ — marching to 1 with the logarithmic correction visible. Everything the cone identification used — the $1/\Delta x^2$ kernel, mutual locality at leading order, the Lorentz-invariant singularity structure — is the m -independent part. The spacetime derivation is safe by the relevance of mass.

3. Causality kept, Huygens lost

Claim. The massive commutator vanishes at spacelike separation and acquires an interior (timelike) tail. **Verification.** From the regulated mode integral: at the spacelike point $(t, x) = (1, 2), \Delta = 1.38 \times 10^{-5}$ — regulator level against interior values of order one; at the interior point $(t, x) = (2, 1), \Delta = 0.332234$ against the closed form $\frac{1}{2} J_0(m\sqrt{3}) = 0.332459$. The audit of axioms is the point: microcausality was the *derived output* of the interior-net paper, and it survives exactly; Huygens sharpness was never an axiom, and its loss is not a wound but a measured tail. The Haag-Kastler net structure of the massive interior stands.

4. The massive wedge is thermal: Bisognano-Wichmann crosses the frontier

Claim. The wedge modular group of the massive theory is the boost flow at $\beta = 2\pi$.

Verification. Along the boost orbit at $\rho = 1$, the Wightman function $W(\tau) = \frac{1}{2\pi} K_0(m s(\tau))$ with the $i\delta$ prescription on Δt (principal branch, $\text{Re } s > 0$, so the K_0 integral converges by construction — a convention-light implementation) was transformed in the boost parameter. The δ -shift contributes the exactly known factor, so the test is $|\tilde{W}(+\omega)/\tilde{W}(-\omega)| = e^{-(2\pi-2\delta)\omega}$.

ω	measured	prediction	rel. dev.
0.25	2.272×10^{-1}	2.131×10^{-1}	6.6%
0.50	4.450×10^{-2}	4.543×10^{-2}	2.1%
0.75	1.011×10^{-2}	9.683×10^{-3}	4.4%
1.00	1.980×10^{-3}	2.064×10^{-3}	4.1%
1.25	4.499×10^{-4}	4.399×10^{-4}	2.3%
1.50	8.800×10^{-5}	9.376×10^{-5}	6.1%

Three and a half decades of exponential law, no fitted parameters, quadrature-limited scatter. The consequence list is the gravity thread's survival: the wedge modular Hamiltonians exist and are the boost generators (Bisognano–Wichmann, valid for the massive field), so the stress-tensor first law, Araki's relative entropy machinery, and the equilibrium-constraint derivation of the linearized Einstein equation are all *mass-robust* — every one of them used only wedge modular structure at its foundation.

5. What is massless-specific, honestly

Three structures do not cross. (i) Generator-diagonal cone data: the massive field's characteristic data are not generator-diagonal; the cone is the derivation scaffold, not the massive theory's home — the massive theory lives in the interior it delivered. (ii) The conformal interval and diamond flows: Hislop–Longo geometry is massless; the massive diamond modular Hamiltonian is a known-hard open structure, cited to Longo–Morsella and left at the frontier. (iii) Exact scale symmetry: mass is a relevant coupling with a dilation weight, converting scale symmetry into renormalization-group motion — the same dimension bookkeeping the covariance audit established for the tower's μ , now appearing in the infrared where, unlike a deformed kinematics, a state-stratum scale is precisely what is allowed.

6. The architecture, stated

Spacetime is derived in the ultraviolet by the massless scaffold; mass is installed in the infrared by the marriage of chiralities and its condensate; the seam between them is consistent because relevance means ultraviolet invisibility — and the seam has now been *tested* at both ends: the singularity coincidence (§2) where the derivation worked, and the thermal wedge (§4) where the gravity thread continues. The massive interior net exists, its axioms audited, its modular backbone verified.

7. Scope

Five items. (i) The detailed-balance check is quadrature-limited at the few-percent level over 3.5 decades; a higher-precision pass is a numerical refinement, not a physics gap. (ii) Massive diamonds remain open and cited. (iii) The interacting massive theory belongs to

the perturbative-algebraic road of the mediator paper. (iv) The multi-component graded massive net — the marriage matrices of the mass paper installed componentwise — is the arena of the road map's Step B, the graded gap system with the rigid channel structure. (v) Step B is next.

References

- Companion papers: *Mass as Marriage: The Chirality Pairing, the Unit Condensate, and the Klein Anatomy of Mass Matrices*; *The Rigid Grading: Layer Inequivalence, the Cross-Split Marriage, and the Algebra's Pre-Order on Mass Channels*; *The Interior Net: Local Algebras from the Cone and Microcausality as Output*; *The Assembly: Null Lines Through a Point, the Light-Cone Identification, and Lorentz Symmetry as Output*; *The First Law on the Net: The Stress Tensor, the Wedge Modular Hamiltonian, and the Relative Entropy of Coherent Excitations*; *The Equilibrium Constraint: One Killing Vector's Two Faces, the Geometric Lemma, and the Linearized Einstein Equation as an Equation of State*; *The Covariance Audit: Why the Native Deformation Cannot Ride the Boosts, and Where Interactions Must Enter*.
- Bisognano, J. J. & Wichmann, E. H., "On the duality condition for a Hermitian scalar field," *J. Math. Phys.* 16 (1975), 985–1007; "On the duality condition for quantum fields," *J. Math. Phys.* 17 (1976), 303–321.
- Longo, R. & Morsella, G., "The massive modular Hamiltonian," arXiv:2012.00565.

Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.